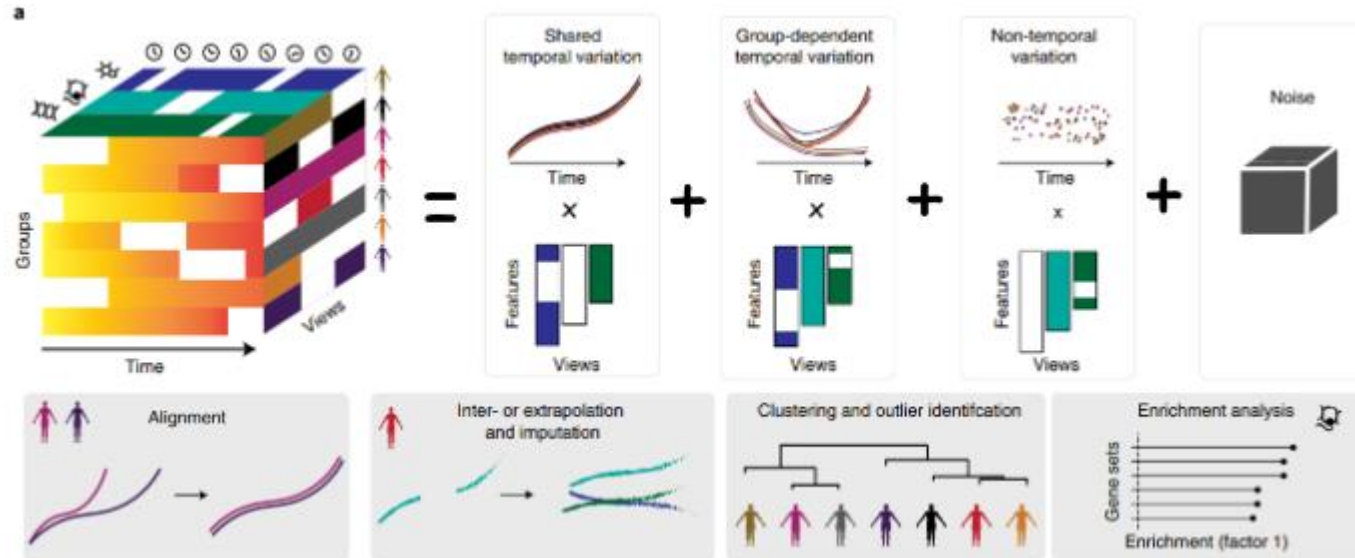


Factor analysis for spatial transcriptomics data

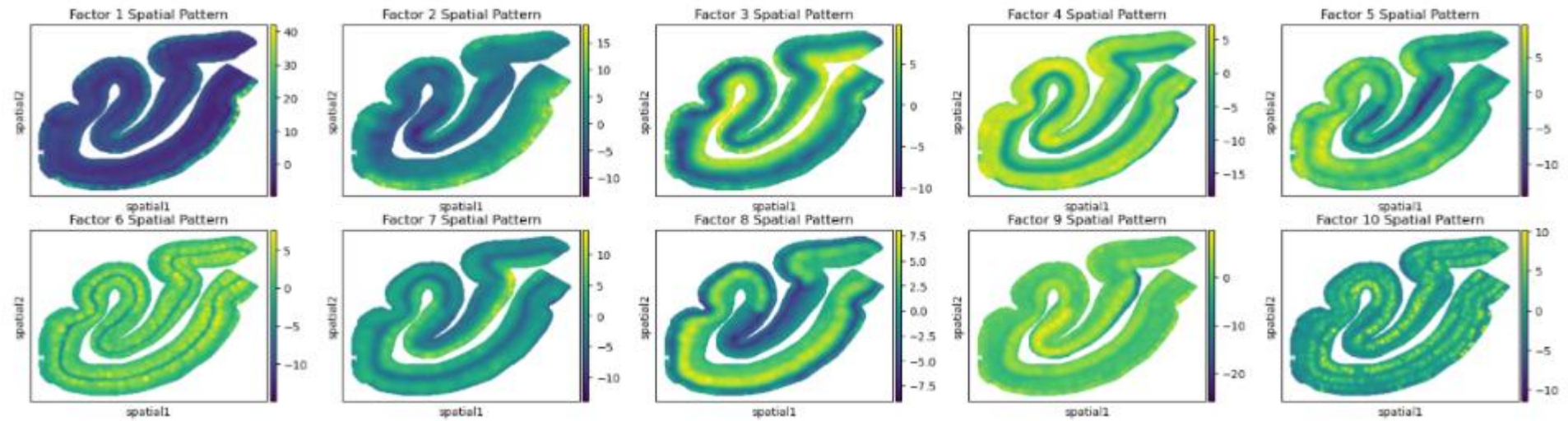
MEFISTO

MEFISTO: Main ideas



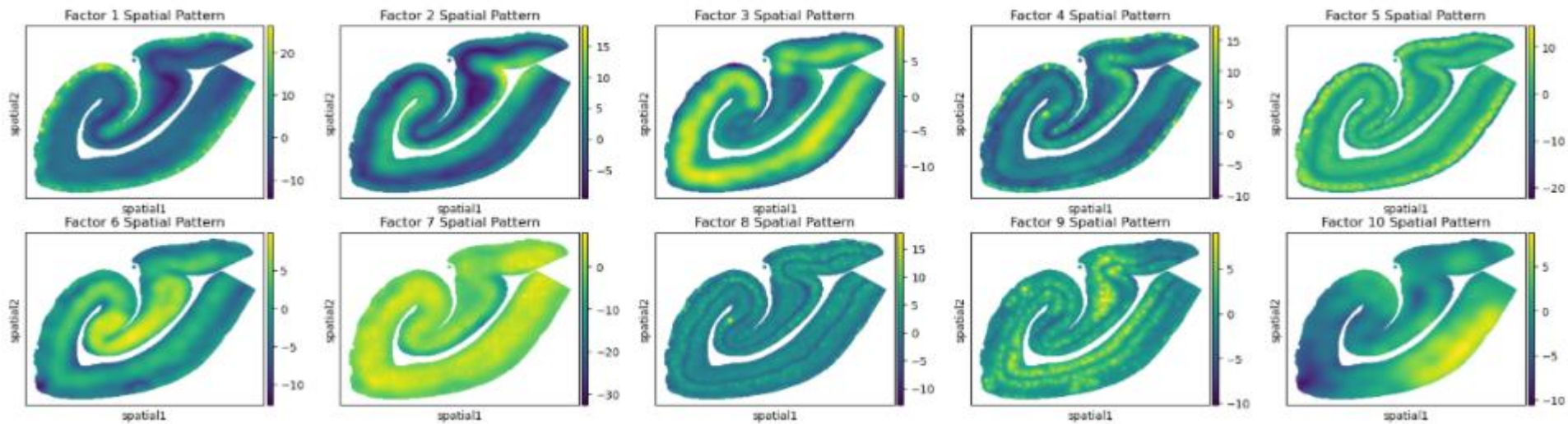
Views, groups and covariate(time or space)

P0, slice 1, 10 factors



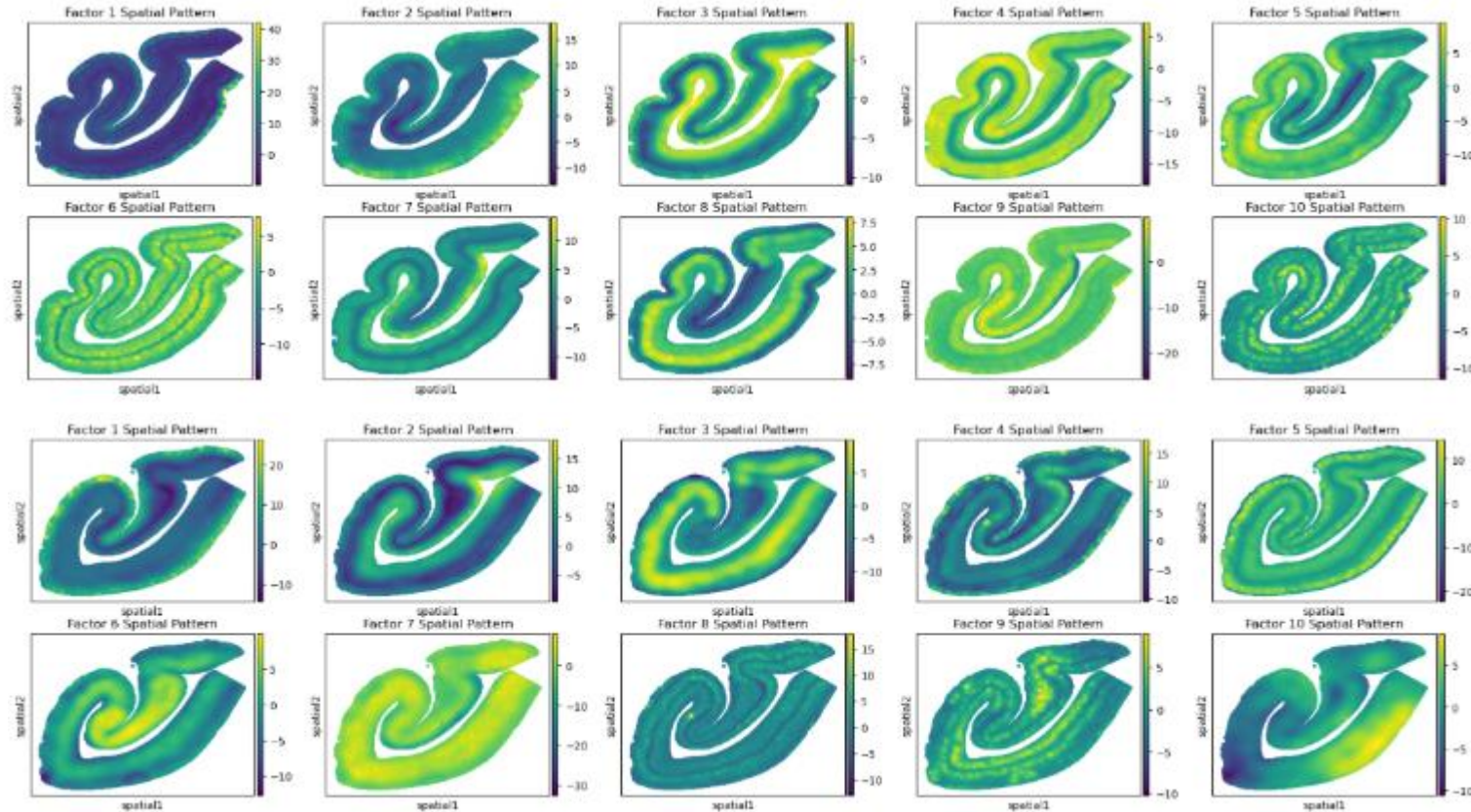
T1089_MT66_P0_Visual_Sub-CTX_b100_filtered.h5ad

P0, slice 2, 10 factors



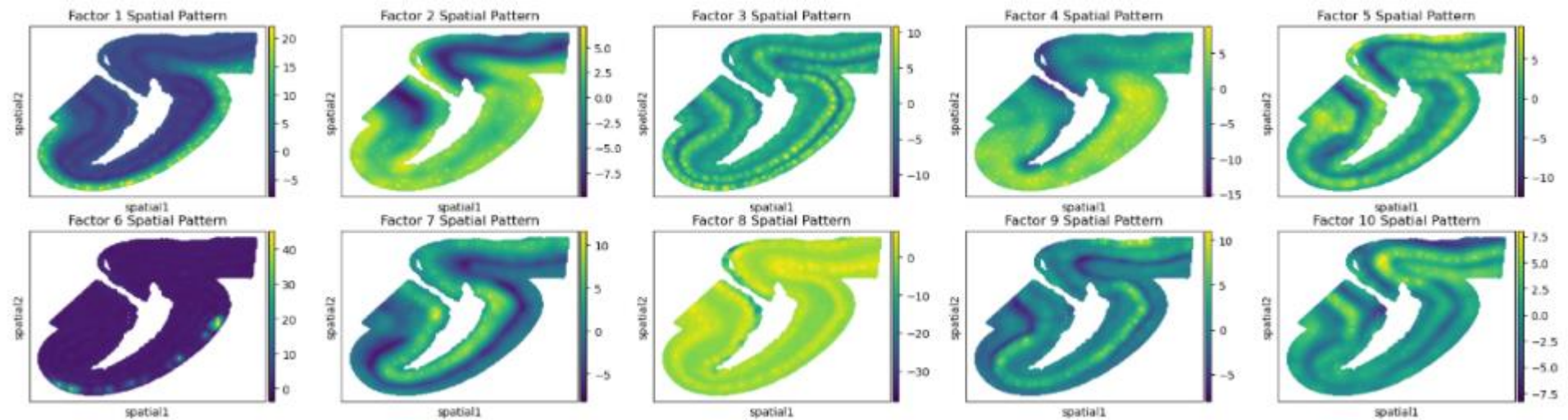
T1090_MT65_P0_Visual_Sub-CTX_b100_filtered.h5ad

Comparison of slices 1 and 2 of P0



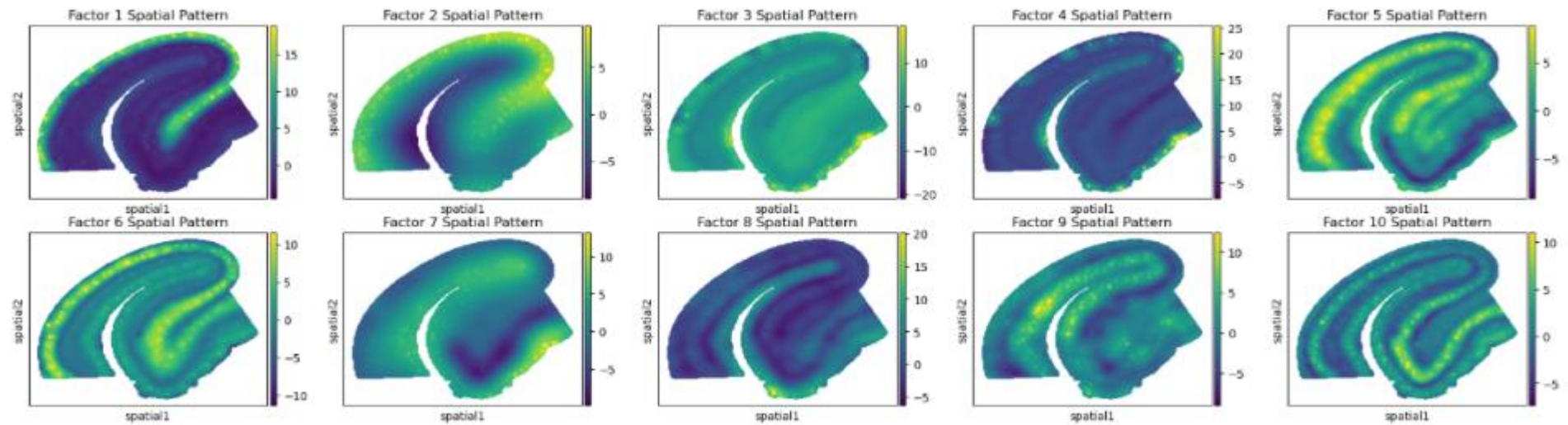
Factors are not consistent across slices. Factors probably align with layers not regions

P32, slice 1, 10 factors



T1091_MT64_P32_Visual_Sub-CTX_V12_b100_filtered.h5ad

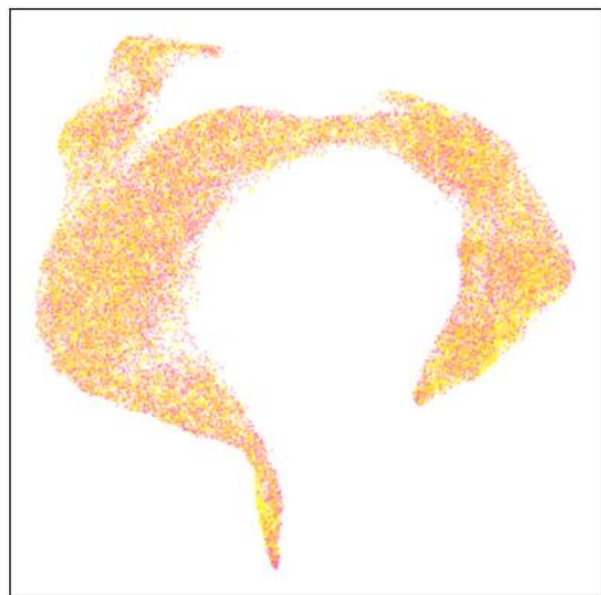
P32, slice 2, 10 factors



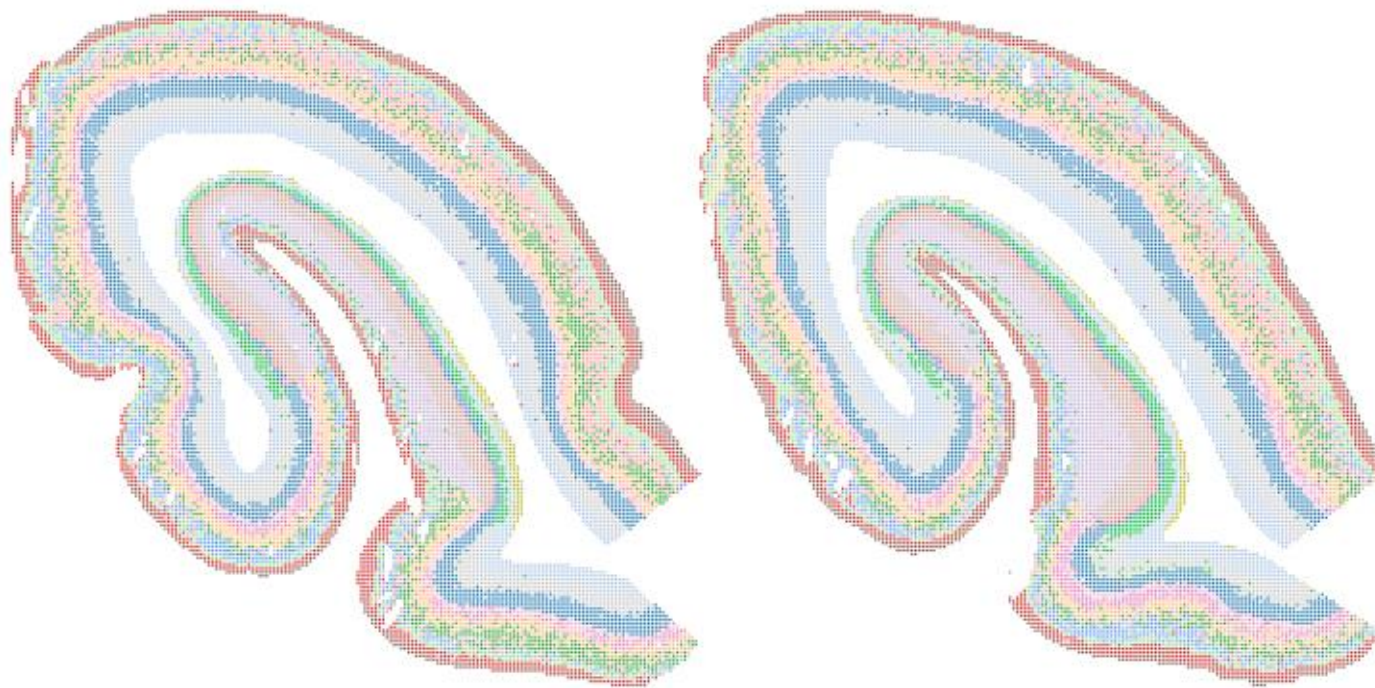
T1092_MT63_P32_Visual_Sub-CTX_V12_b100_filtered.h5ad

INSPIRE

Two slices of the same age and similar coordinate

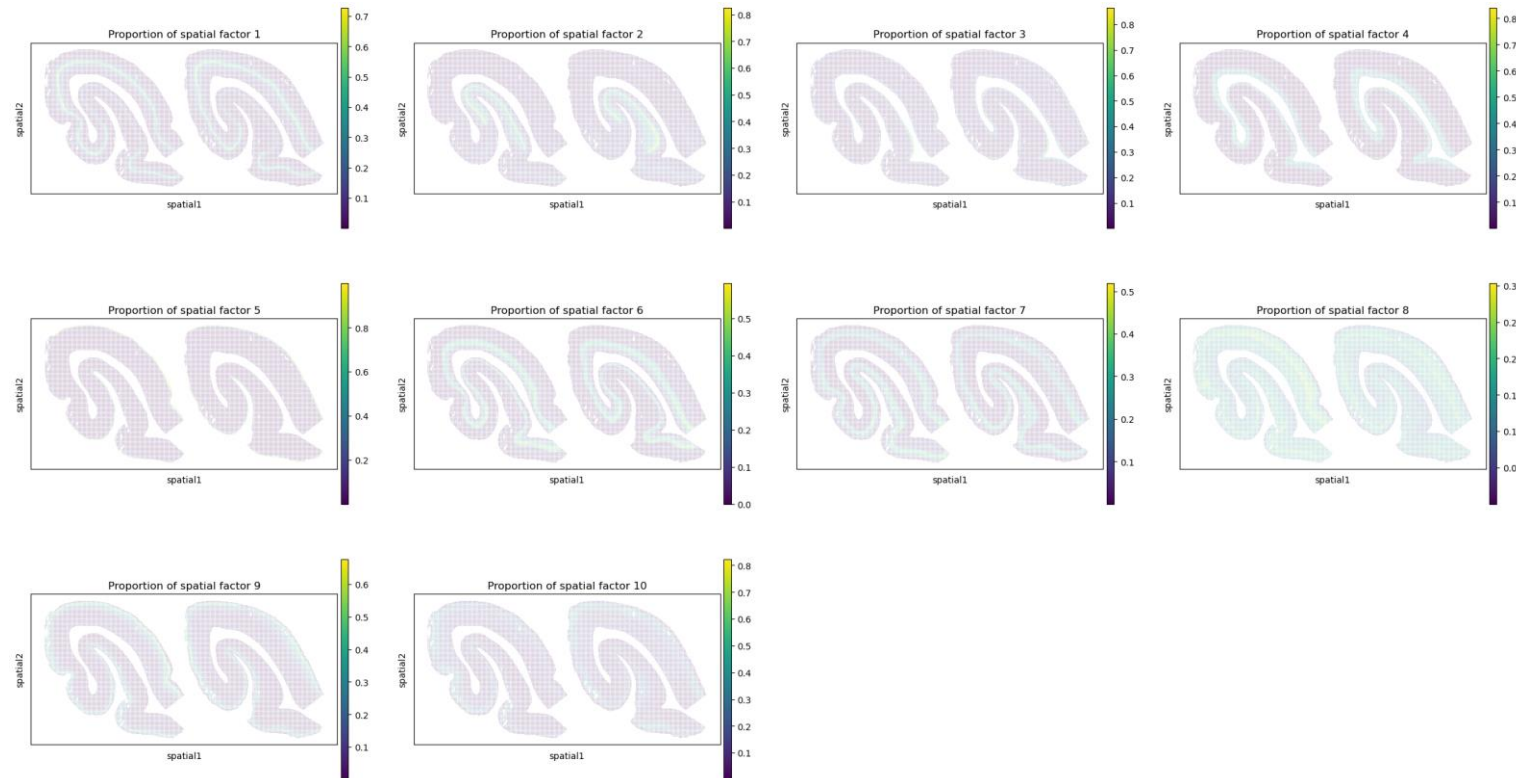


Slices are aligned in the latent space



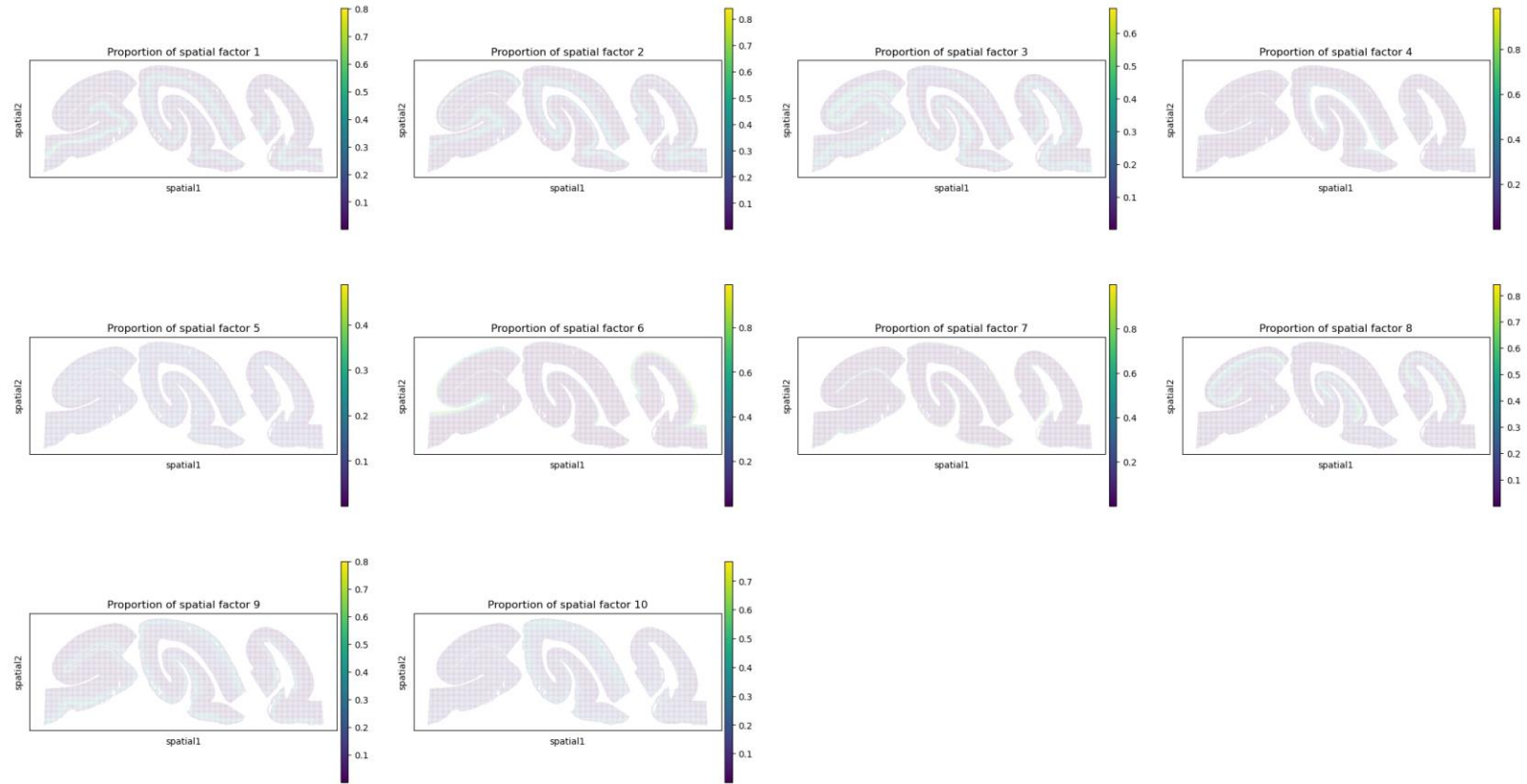
INSPIRE computes factors consistent between slices

10 factors for two slices



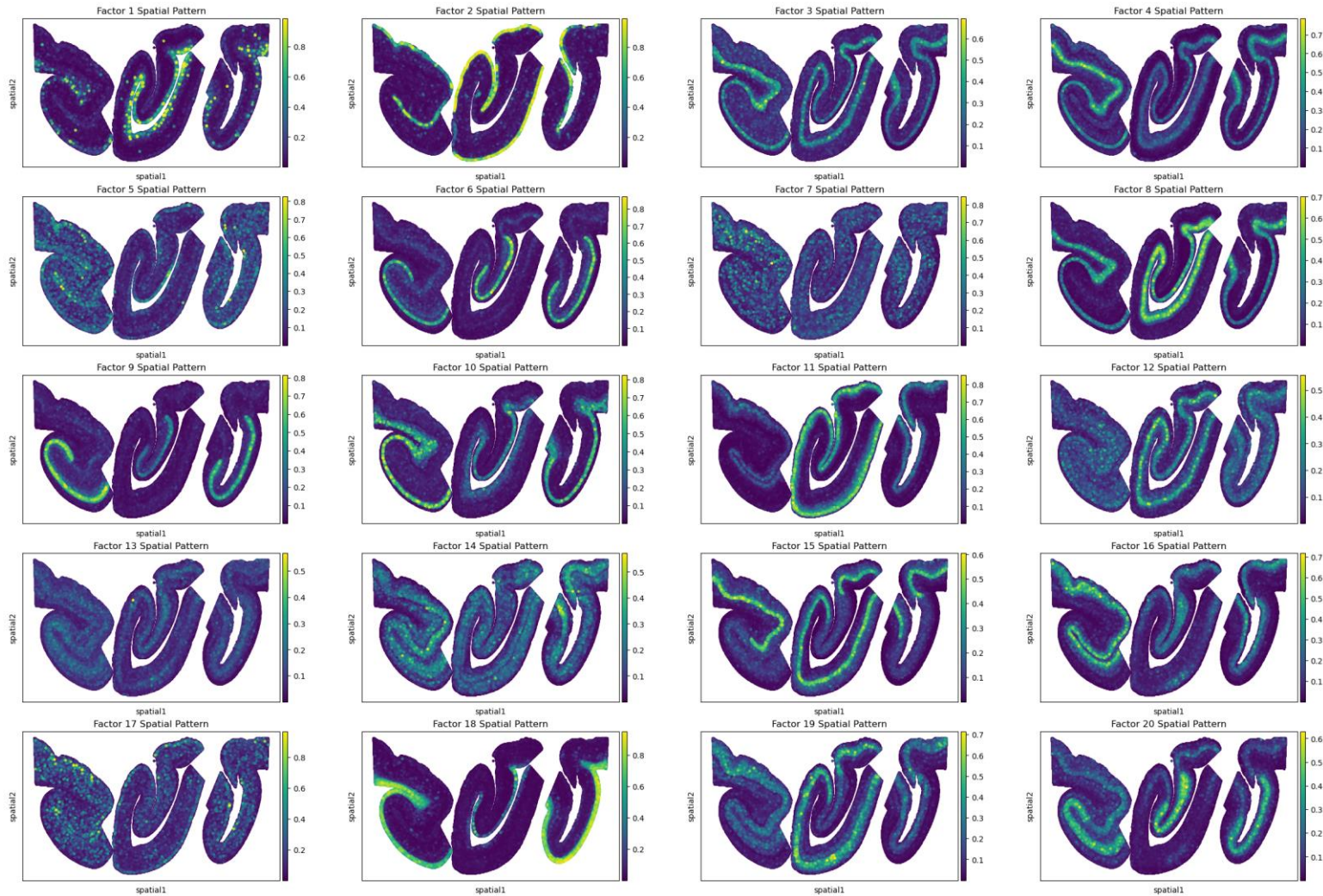
P0 slices, all factors

10 factors for 3 slices

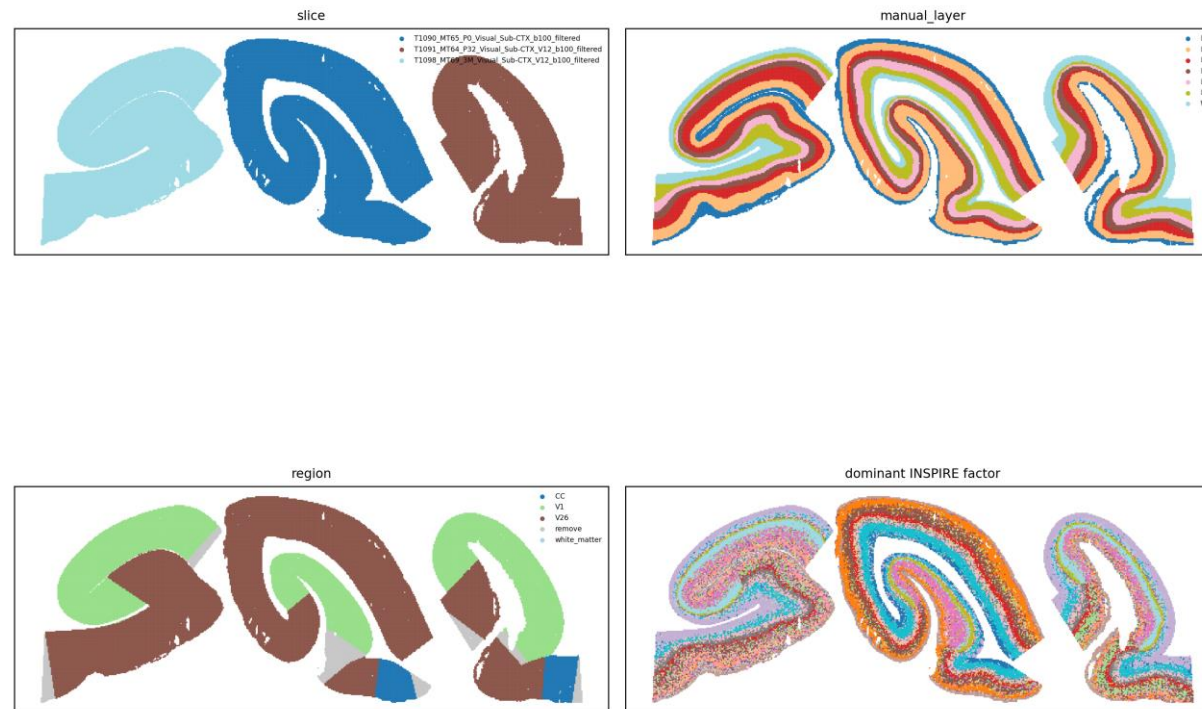


P0, P32 and 3M slices; all factors

20 factors for 3 slices

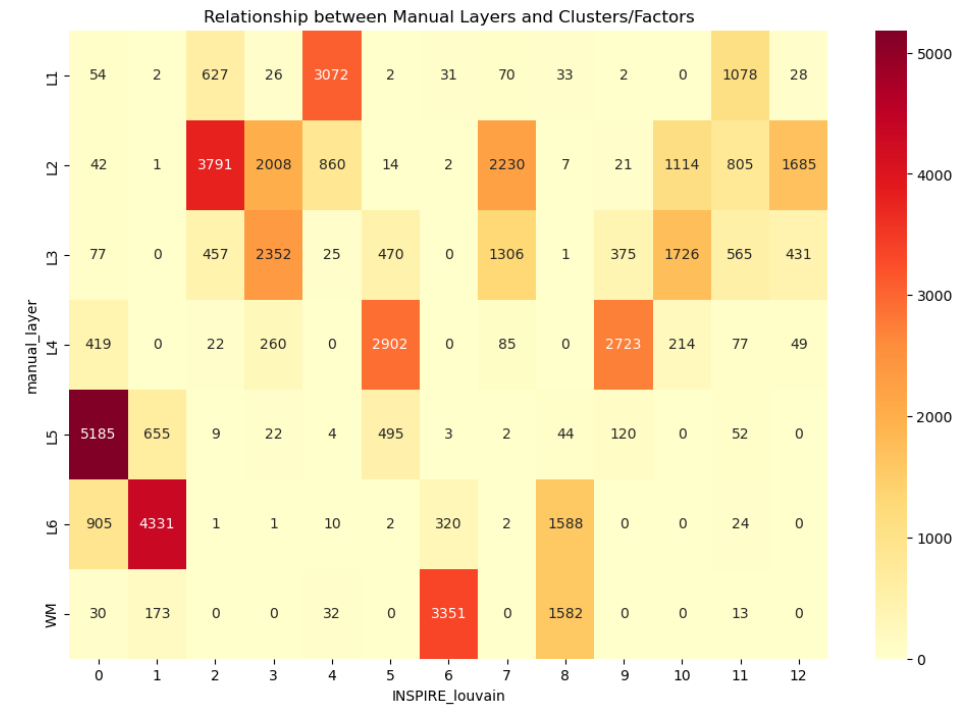
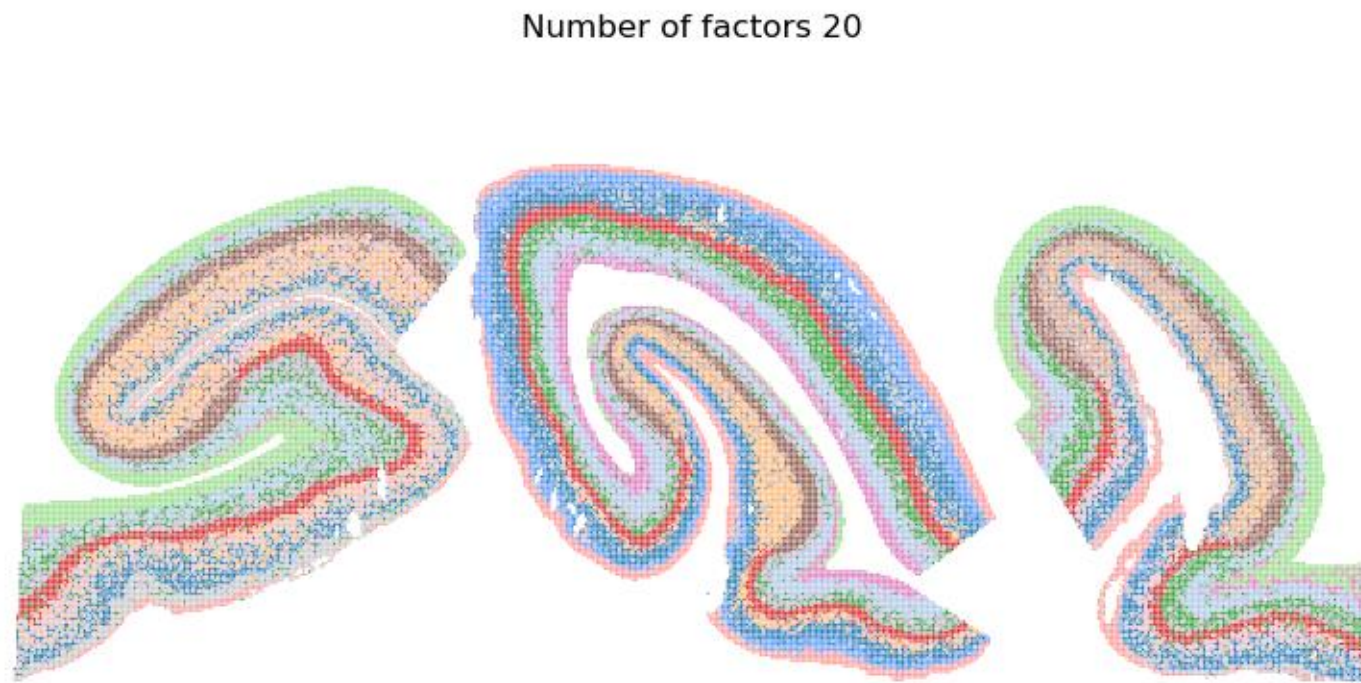


Dominant factor and manual annotation for 20 INSPIRE factors on P0, P32, 3M



Analysis was done for 10, 20,40,60,80,100 factors. Increasing number of factors leads to noisy factors and dose not allow to be closer to distinguishing regions

Louvain clusters based on factors from INSPIRE on 3 slices with 20 factors



Right picture shows that computed factors correspond to layers

Conclusions

- MEFISTO gives good factors, but models trained on individual slices does not give consistent factors. Training the model on several slices uses less efficient algorithm and therefore takes more time, but can be computed. Also spatial coordinates are considered as covariates, time is taken into account as group. Disadvantages – compute time, no GPU optimization, spatial coordinates of different slices are not aligned by MEFISTO. It is possible to use before MEFISTO tools designed for alignment, but they need MRI or ST template which we do not have.
- INSPIRE potentially is well suited for computing consistent between slices factors that well correspond to layers. But regions are not distinguishable, because probably difference between layers is bigger and those directions correspond to higher variance